REVIEW 2ND QUARTER

CALCIUM PHOSPHATE REACTS WITH SILICON DIOXIDE AND CARBON PRODUCING CALCIUM SILICATE, CARBON DIOXIDE, AND PHOSPHORUS. WHAT IS THE BALANCED EQUATION FOR THIS REACTION?



SILVER NITRATE REACTS WITH MAGNESIUM CHLORIDE TO PRODUCE SILVER CHLORIDE AND MAGNESIUM NITRATE. IF 305 GRAMS OF SILVER NITRATE ARE REACTED IN AN EXCESS OF MAGNESIUM CHLORIDE PRODUCING 23.7 GRAMS OF MAGNESIUM NITRATE, HOW MUCH IS THE PERCENT YIELD?



A slightly underinflated rubber life raft left in bright, summer sunlight swells up. You should not overinflate your life raft when your ship goes down in tropical waters because the gas particles' average kinetic energy increases and the gas's volume increases. Which GAS law explains this phenomenon?

Answer: Charles's Law



A student must solve the following problem: Neon gas is confined to 2.00 L at standard temperature and pressure.
Find the temperature when the volume and pressure are doubled. Which gas law should the student use to solve
this problem? answer: Combined Gas Law



The table shows data for the temperature, pressure, and volume of four gas samples. Which two gas samples contain the same number of molecules? answer: II and IV



Data for Four Gases

Gas Sample	Temperature (K)	Pressure (atm)	Volume (L)
I	600.	2.0	5.0
II	300.	1.0	10.0
III	600.	3.0	5.0
IV	300.	1.0	10.0

Which phrase describes the motion and attractive forces of ideal gas particles?
A. random straight-line motion and strong attractive forces
B. RANDOM STRAIGHT-LINE MOTION AND NO ATTRACTIVE FORCES
C. random curved-line motion and no attractive forces
D. random curved-line motion and strong attractive forces



In a gaseous mixture at 20 °C, the partial pressure of the components are: Hydrogen, 150 torr; Carbon Dioxide, 200 torr; Methane, 320 torr; Ethylene, 105 torr. What is the total pressure of the mixture?
Answer: Ptotal = PH2 +PCO2 + PCH4 + PC2H4
Ptotal = 775 torr



A 3.0 L SAMPLE AT 25°C AND 2.00 ATM OF PRESSURE CONTAINS 0.52 MOLE OF A GAS. IF AN ADDITIONAL 0.35 MOLE OF GAS AT THE SAME PRESSURE AND TEMPERATURE IS ADDED, WHAT IS THE FINAL TOTAL VOLUME OF THE GAS?



IF THE TEMPERATURE OF A CONFINED GAS SAMPLE IS DOUBLED, WHILE THE VOLUME IS HELD CONSTANT, WHAT WILL HAPPEN TO THE PRESSURE?
(A) IT WILL DOUBLE (GAY-LUSSAC'S LAW)
(B) IT WILL BE FOUR TIMES AS LARGE.
(C) IT WILL BE HALF AS LARGE
(D) CANNOT BE DETERMINED WITH INFORMATION HERE



WHICH OF THE FOLLOWING STATEMENTS DOES NOT DESCRIBE A LIMITING REAGENT?
A. IT DETERMINES THE AMOUNT OF PRODUCT FORMED.
B. IT DETERMINES THE AMOUNT OF REACTANT FORMED.
C. IT IS THE REACTANT THAT IS FULLY CONSUMED IN A GIVEN REACTION.
D. IT IS THE REACTANT THAT PRODUCES THE LEAST AMOUNT OF PRODUCT

WHEN A SAMPLE OF OXYGEN GAS IN A CLOSED CONTAINER OF CONSTANT VOLUME IS HEATED UNTIL ITS ABSOLUTE TEMPERATURE IS DOUBLED, WHICH OF THE FOLLOWING IS ALSO DOUBLED?
A. THE DENSITY OF THE GAS
B. THE PRESSURE OF THE GAS
C. THE NUMBER OF MOLECULES PER CM^3
D. THE AVERAGE VELOCITY OF THE GAS MOLECULES

IN A CONTAINER, GAS MOLECULES WILL OCCUPY THE AVAILABLE SPACE. ACCORDING TO THE KINETIC MOLECULAR THEORY, WHAT HAPPENS TO THE GAS MOLECULES WHEN THEY COLLIDE?
A. THE COLLISIONS ARE SO HARD THE GAS MOLECULES BREAK APART.
B. THE COLLISIONS ARE PERFECTLY ELASTIC AND THERE IS NO NET LOSS OF KINETIC ENERGY.
C. THE COLLISIONS ARE INELASTIC AND THERE IS LESS KINETIC ENERGY AFTER THE COLLISION.
D. THE COLLISIONS ARE PERFECTLY ELASTIC AND THERE IS MORE KINETIC ENERGY AFTER THE COLLISION.

WHICH GASES WILL HAVE THE GREATEST EFFUSION RATE AT A GIVEN TEMPERATURE?
A. Ar
B. CH_4
C. HCl
D. HBr

WHICH OF THE FOLLOWING STATEMENTS IS INCONSISTENT WITH THE KINETIC-MOLECULAR THEORY OF GASES?
A. THE AVERAGE KINETIC ENERGIES OF DIFFERENT GASES DIFFER AT THE SAME TEMPERATURE.
B. THE THEORY EXPLAINS MOST OF THE OBSERVED BEHAVIOR OF GASES AT ORDINARY TEMPERATURES AND PRESSURES.
C. THERE IS NO NET GAIN OR LOSS OF THE TOTAL KINETIC (TRANSLATIONAL) ENERGY IN COLLISIONS BETWEEN GAS MOLECULES.
D. THE ACTUAL VOLUME OF THE GAS MOLECULES THEMSELVES IS VERY SMALL COMPARED TO THE VOLUME OCCUPIED BY THE GAS AT ORDINARY TEMPERATURES AND PRESSURE

WHY DOES THE PRESSURE INSIDE A CONTAINER OF GAS INCREASE IF MORE GAS IS ADDED TO THE CONTAINER?
A. THERE IS A DECREASE IN THE VOLUME OF THE GAS.
B. THERE IS AN INCREASE IN THE TEMPERATURE OF THE GAS
C. THERE IS AN INCREASE IN THE NUMBER OF COLLISIONS BETWEEN PARTICLES AND THE WALLS OF THE CONTAINER.
D. THERE IS AN INCREASE IN THE FORCE OF THE COLLISIONS BETWEEN THE PARTICLES AND THE WALLS OF THE CONTAINER.

THREE 1.0 LITER FLASKS ARE FILLED WITH OXYGEN, HYDROGEN, AND NEON GAS, RESPECTIVELY, AT STP. WHICH OF THE FOLLOWING STATEMENTS IS TRUE?
A. GASES IN EACH CONTAINER HAVE THE SAME DENSITY.
B. THERE ARE TWICE AS MANY O2 AND H2 MOLECULES AS N ATOMS.
C. EACH FLASK HAS THE SAME NUMBER OF GAS MOLECULES.
D. THE VELOCITY OF THE GAS MOLECULES IS THE SAME IN EACH FLASK.

A 250 ml sample of oxygen is collected over water at 25°C and 760 torr pressure. What is the pressure of the dry gas alone? (Vapor pressure of water at 25°C is equal to 23.8 torr)
Answer: Total pressure of sample is 760 torr and for the pressure of water is 23.8 torr
Recall: Total pressure = Pa (water) + Pb (dry gas)
760 torr = 23.8 torr + Pdry gas
Pdry gas = 760 torr - 23.8 torr
Pdry gas = 736.2 torr

Potassium perchlorate decomposes producing potassium chloride and oxygen.

$$2KClO_3 \rightarrow 2KCl + 3O_2$$

[K= 19 ; O=16 ; Cl= 35]
What is the volume of oxygen gas at 27°C and 0.821 atm?
answer: 90.0 L use PV=nRT to calculate the answer, moles of oxygen is based on the Balanced Chemical Eq.

Under which conditions of pressure and temperature is a real gas most like an ideal gas?
A. low pressure and low-temperature
B. low pressure and high-temperature
C. high pressure and low temperature
D. high pressure and high temperature

A SAMPLE OF A CERTAIN GAS HAS A VOLUME OF 415 ML AT 725 MM HG AND 0 OC. WHAT WOULD BE THE VOLUME OF THIS SAME SAMPLE OF GAS IF IT WERE MEASURED AT STP?

For the given reaction,
$2MnO2 + O2 + Cl2 \quad 2KMnO4 + 2KCl + 2H2O$,
there is 100 g of each reactant available.
Which reagent is the limiting reagent?
[Molar Masses: Mn 55; O 16; K 39; Cl 35]
Answer: MnO2 (which has the largest molar mass and has the same coeff. with the products)

Which of the following statements is TRUE about ideal gases?
A. They have a volume
B. They are not attracted to each other
C. An ideal gas conforms to the ideal gas law only at STP
D. A real gas is an ideal gas at all temperatures and pressures

A glass box holds equal amounts of hydrogen, nitrogen, oxygen, and bromine. The gases are allowed to exit the container through a tiny hole. Which gas will exit the hole the fastest?
A. Bromine
B. Hydrogen
C. Nitrogen D. They all exit at the same.
Answer: Calculate the mass of the gas, the lighter the gas the fastest it will exit. Letter B

Use the following reaction to answer the next few questions:

$$2\ C_8H_{18} + 25\ O_2 \rightarrow 16\ CO_2 + 18\ H_2O$$

The above reaction is the reaction between gasoline (octane) and oxygen inside automobile engines.If you burned one gallon of gas (C_8H_{18}) (approximately 4000 grams), how many liters of carbon dioxide would be produced at a temperature of 21°C and a pressure of 1.00 atm?

Solution-PV=nRT, where can be calculated by mass/molar mass

Sodium chloride reacts with plumbous chloride producing sodium nitrate and plumbic chloride.

$$2NaCl + Pb(NO_3)_2 \quad 2NaNO_3 + PbCl_3$$

How many grams of lead (II) chloride is produced from the reaction of 15.3 g of NaCl and 60.8 g of $Pb(NO_3)_2$?

[Na=23 ; Cl=35 ; Pb = 208 ; O=16]